The Vanishing Assembler Workforce: A Hidden Risk to Global Financial Stability

Neeraj Aggarwal 6 min read · Feb 16, 2026

The Vanishing Assembler Workforce: A Hidden Risk to Global Financial Stability

Introduction: The Legacy Skills Crisis No One Wants to Own

Across banking, aviation, government, and global payments, the world's most mission-critical systems still rely on technologies that predate the internet. Assembler, COBOL, and Transaction Processing Facility (TPF) systems quietly power trillions of dollars in daily transactions. Yet the number of professionals capable of maintaining, securing, or modernizing these systems is collapsing at a pace that now threatens operational continuity.

This is not an abstract concern. The U.S. Government Accountability Office (GAO) recently released a 2025 report identifying **11 high-risk federal systems** urgently requiring modernization, many of which run on **Assembler and COBOL** [1]. The GAO explicitly warns of a **"dwindling number of people available with the skills needed to support them"** [1]. Treasury, IRS, and Social Security systems — the backbone of U.S. economic infrastructure — are among the most vulnerable.

The private sector faces the same challenge. Banks, airlines, and payment networks continue to depend on TPF and Assembler-based platforms because they deliver unmatched throughput, reliability, and latency. But the workforce that built and maintained these systems is retiring, and the next generation is not being trained to replace them.

1. Why Assembler Still Matters in 2026

Assembler is often dismissed as "legacy," but in reality it is **specialized infrastructure technology** — the digital equivalent of the power grid. It is invisible, indispensable, and extremely difficult to replace.

1.1 The workloads that refuse to move

IBM's modernization analysis highlights a pattern that many CIOs privately acknowledge:
Attempts to rewrite high-volume transactional systems into cloud-native architectures often fail to meet performance requirements [2].

IBM cites cases where:

- A cloud-native rewrite delivered **only one-third** of the throughput of the original IBM Z/TPF system [2].
- A financial institution spent years migrating to a modern platform, only to **abandon the new system** and revert to TPF because the cloud architecture could not maintain SLAs [2].
- Rewrites became **9× more expensive** than anticipated due to hidden complexity and undocumented business logic [2].

These failures are not due to poor engineering. They stem from the fact that Assembler and TPF were designed for **microsecond-level performance**, deterministic execution, and extreme concurrency — characteristics that modern distributed systems struggle to replicate.

1.2 The industries that depend on it

Assembler remains foundational in:

- **Banking:** Core transaction processing, card authorization, settlement
- **Aviation:** Reservation systems, ticketing, real-time seat inventory
- **Payments:** High-volume switching, fraud checks, clearing
- **Government:** Tax processing, benefits distribution, identity systems

These are not niche workloads. They are the systems that keep economies functioning.

2. The Shrinking Talent Pool: A Structural Workforce Collapse

The Assembler workforce is not just shrinking — it is collapsing.

2.1 Retirement cliff

Most Assembler engineers are in their late 50s to early 70s. Many were hired in the 1970s–1990s and have already retired or are planning to retire within the next five years.

2.2 No academic pipeline

Universities stopped teaching Assembler and TPF decades ago. Computer science programs focus on Python, Java, cloud-native architectures, and AI — none of which prepare graduates to maintain low-level transactional systems.

2.3 Corporate training has disappeared

In the 1990s and early 2000s, banks and airlines ran internal academies for mainframe and Assembler training. Most of these programs were shut down during cost-optimization cycles.

2.4 Modernization programs underestimate skill depth

Organizations often assume that:

- “Assembler can be replaced by Java.”
- “We can hire contractors.”

- "We can rewrite the system in the cloud."

These assumptions repeatedly prove false. The GAO report shows that even federal agencies with billion-dollar budgets struggle to find the talent needed to maintain or modernize legacy systems [1].

3. The Modernization Bottleneck: Why Rewrites Fail

The Forbes analysis of IRS modernization illustrates a broader truth: Modernization is not a technology problem — it is a **knowledge continuity problem** [3].

3.1 Legacy logic is deeply embedded

Assembler systems often contain:

- 40+ years of business rules
- Hard-coded regulatory logic
- Performance optimizations that are undocumented
- Interdependencies with dozens of downstream systems

Rewriting these systems requires not just coding skill, but **institutional memory** — something that is rapidly disappearing.

3.2 Documentation is incomplete or outdated

Many Assembler systems were built before modern documentation practices existed. Engineers relied on tribal knowledge, whiteboards, and in-person collaboration. When these engineers retire, the logic retires with them.

3.3 Modern platforms cannot replicate deterministic performance

Cloud-native systems introduce:

- Network latency
- Distributed concurrency
- Non-deterministic execution
- Higher operational overhead

For workloads requiring microsecond-level response times, these trade-offs are unacceptable.

3.4 Regulatory pressure is increasing

The Bank Competition Modernization Act debates highlight a growing tension:
Regulatory modernization is accelerating, but technical modernization is lagging — widening the compliance gap [4].

4. The GAO Warning: A National-Level Risk

The GAO's 2025 report is one of the strongest public acknowledgments of the legacy skills crisis.

Key findings include:

- **11 critical federal systems** urgently require modernization [1].
- **8 of these systems** rely on legacy languages such as COBOL and Assembly [1].

- Agencies lack complete modernization plans, increasing the risk of cost overruns and failures [1].
- The workforce capable of maintaining these systems is **"dwindling."**
- Some systems are **over 50 years old** and still process billions of dollars in transactions.

This is not just a government problem. It is a **global economic risk.**

5. Banking's Hidden Dependency: TPF and Assembler

Banks rarely discuss their reliance on Assembler publicly, but the dependency is deep and structural.

5.1 Real-time transaction processing

Card authorization, ATM withdrawals, and interbank transfers often run on TPF-based systems. These systems process:

- Tens of thousands of transactions per second
- With sub-millisecond latency
- At near-perfect uptime

No modern platform matches this combination of speed, reliability, and cost efficiency.

5.2 The modernization paradox

Banks want to modernize, but:

- They cannot risk downtime.
- They cannot replicate TPF performance.
- They cannot find enough Assembler engineers to support a rewrite.

This creates a modernization paradox:
The systems most in need of modernization are the least replaceable.

6. What the Industry Must Do Now

The Assembler talent shortage is not reversible, but it is manageable — if organizations act now.

6.1 Treat Assembler as a strategic capability

Stop viewing it as "legacy."
Start viewing it as **critical infrastructure**.

6.2 Build internal academies

Banks, airlines, and government agencies must re-establish structured training programs.
This is the only sustainable way to rebuild capability.

6.3 Modernize selectively

Not every system should be rewritten.
Hybrid modernization — isolating components, wrapping APIs, and incrementally refactoring — is more realistic.

6.4 Capture institutional knowledge before it disappears

Organizations must:

- Document business logic
- Record expert walkthroughs
- Build knowledge repositories
- Pair retiring engineers with new hires

6.5 Partner with universities

Introduce courses on:

- Systems programming
- Mainframe architecture
- High-volume transaction processing

6.6 Mandate modernization plans

The GAO recommends requiring agencies to document modernization strategies [1].
Banks should adopt the same discipline.

Conclusion: The Time to Act Is Now

The scarcity of Assembler professionals is no longer a niche workforce issue — it is a systemic risk to global financial stability. The world's most critical systems depend on a shrinking pool of experts, and modernization efforts repeatedly fail because they underestimate the depth of legacy knowledge embedded in these platforms.

The organizations that succeed will be those that:

- Invest in capability-building
- Modernize pragmatically

- Preserve institutional knowledge
- Treat Assembler as a strategic asset

The rest will face escalating operational risk, rising modernization costs, and an eventual crisis of continuity.

The Assembler workforce is vanishing.
The systems it supports are not.
The time to act is now.

References

[1] U.S. Government Accountability Office (GAO).
Federal Agencies Need to Address Aging IT Systems and Workforce Risks.
GAO-25-107795, 2025.
https://www.gao.gov/products/gao-25-107795

[2] IBM.
Busting the Biggest Myths About Modernization. IBM Think Insights.
https://www.ibm.com/think/insights/busting-the-biggest-myths-about-modernization (ibm.com)

[3] Leahey, Andrew.
"From COBOL and Assembly to Java: Modernizing the IRS." *Forbes,* September 6, 2024.
https://www.forbes.com/sites/andrewleahey/2024/09/06/from-cobol-and-assembly-to-java-modernizing-the-irs/

[4] American Action Forum.
The Bank Competition Modernization Act: Some Good, Some Not-So-Good.
https://www.americanactionforum.org/insight/the-bank-competition-modernization-act-some-good-some-not-so-good/

[5] Quora Discussion.
"How is the future of Transaction Processing Facility (TPF) both in India and globally?"
https://www.quora.com/How-is-the-future-of-Transaction-Processing-Facility-TPF-both-in-India-and-globally

About Author

Neeraj Aggarwal is a recognized modernization leader with deep expertise in AI, core banking, and legacy transformation. He advises financial

institutions on building resilient, data-driven architectures and contributes thought leadership across industry forums and publications.

Written by Neeraj Aggarwal

Edit profile

2 followers · 2 following

Senior Technical Program Manager with more than 2 decades in banking modernization. Writing about NLP, automation and the future of financial systems.

No responses yet

Neeraj Aggarwal

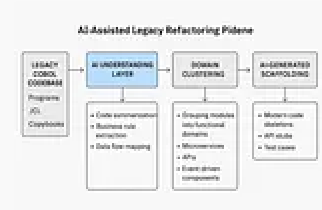
What are your thoughts?

More from Neeraj Aggarwal

Neeraj Aggarwal

AI for Legacy Code Refactoring: From COBOL to Cloud-Native

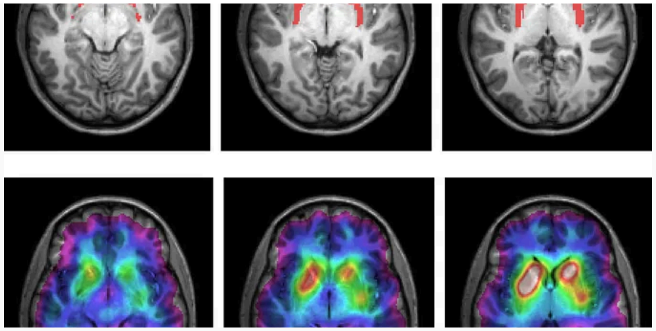
Introduction: From monoliths to cloud-native, with a 1960s core



Dec 30, 2025 1 1

See all from Neeraj Aggarwal

Recommended from Medium



In Write A Catalyst by Dr. Patricia Schmidt

As a Neuroscientist, I Quit These 5 Morning Habits That Destroy You…

Most people do #1 within 10 minutes of waking (and it sabotages your entire day)

Jan 14 37K 671

In CodeX by MayhemCode

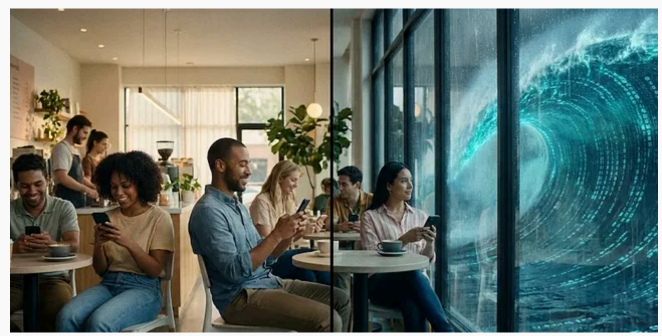
Why Thousands Are Buying Mac Minis to Escape Issues with Big…

Something strange happened in early 2026. Apple stores started running low on Mac…

Feb 15 5K 83

In Realworld AI Use Cases by Chris Dunlop

Senior Developers — The World Owes You an Apology



In Activated Thinker by Shane Collins

Why the Smartest People in Tech Are Quietly Panicking Right Now

Why senior developers still matter in the age of AI coding tools—and what junior...

Feb 27 · 1.91K · 62

The water is rising fast, and your free version of ChatGPT is hiding the terrifying,...

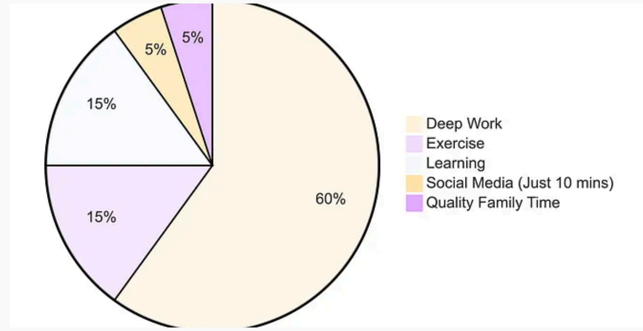
Feb 13 · 13.8K · 573

In GitBit by John Gruber

Websites Are Dead. Go Here Instead.

I finally did it. I launched a blog. Then I realized the hard truth.

Feb 10 · 9.2K · 359



In Level Up Coding by Teja Kusireddy

I Stopped Using ChatGPT for 30 Days. What Happened to My Brai...

91% of you will abandon 2026 resolutions by January 10th. Here's how to be in the 9% who...

Dec 28, 2025 · 10.6K · 377

See more recommendations